
Scalable Fully Bayesian Gaussian Process Classification via Accelerated RPCholesky and HMC

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Abstract

Gaussian Process Classification (GPC) provides inherent uncertainty quantification, which gives it an advantage over neural networks in certain fields of application like cancer diagnosis. However, it scales poorly with data sizes due to the $O(N^3)$ complexity of kernel factorization. Traditional approximation approaches (Expectation Propagation, Laplace Approximation, Variational Inference) sacrifice the posterior exactness for scalability. We present a fully Bayesian GPC using accelerated Randomly Pivoted Cholesky (RPCholesky) for kernel factorization, sampling the exact posterior using Hamiltonian Monte Carlo (HMC), and using a pivot-as-inducing-point scheme for predictive calculation. We applied the method on two cancer datasets with both training data and existing labels, **PCam** and **CAMELYON17-WILDS**. While a fully Bayesian approach was infeasible for datasets with size approximately 300k in previous literature, we managed to produce predictive that are better calibrated than traditional alternatives and neural network baselines, while remaining computationally tractable.

1 Introduction

Neural networks have been widely used for cancer diagnosis from histopathology images. Despite high accuracy, neural network can produce overconfident predictions (Guo et al. [2017]), which is a serious limitation in clinical environment where uncertainty quantification is critical. For Cancer histopathology, false negatives can be life threatening and delay treatment, whereas false positives can cause patient distress and unnecessary medical expenses.

Gaussian Process Classification (GPC) provides principled uncertainty estimates. Historically, approximate inference methods have dominated GPC due to the cost of MCMC ([Nickisch and Rasmussen, 2008]). In this paper, we use Markov Chain Monte Carlo (MCMC) to sample the posterior with a new kernel factorization method called accelerated randomly pivoted Cholesky (RPCholesky). Then we compare our low-rank HMC GPC against the stochastic variational Gaussian process (SVGP) and neural-network baselines.

Contributions. We make the following contributions: (1) We propose the first scalable fully Bayesian GPC that combines accelerated RPCholesky [Epperly et al., 2025] with HMC under non-centered parameterization, making MCMC-based GPC tractable at $N \approx 300,000$ training points; (2) We introduce a pivots-as-inducing-points predictive scheme based on previous literature (Quiñero-Candela and Rasmussen [2005], Williams and Seeger [2001]) that maintains consistency between the training model and predictive by using the pivot property of RPCholesky factorization; and (3) The experiments on **PCam** and **CAMELYON17-WILDS** demonstrate that our method produces

substantially better-calibrated metrics than SVGP and neural-network baselines, with the largest differences under hospital-level distribution shift.

2 Related Work

Low-rank kernel approximation. our method is directly inspired by the recent randomized numerical linear algebra discovery on low-rank factorization for semi-positive definite matrices. Chen et al. [2025] introduced randomly pivoted Cholesky (RPCholesky), a variant of partial Cholesky decomposition that approximates PSD matrices with small trace-norm approximation error. Epperly et al. [2025] and accelerate RPCholesky further improved the factorization time by 40 times in theory with an accelerated implementation that combines block RPCholesky and rejection sampling.

Approximate Inference for GP Classification and Stochastic variational Gaussian processes.

A long line of literature on approximate inference was developed for approximating the intractable GPC posterior, compared by Nickisch and Rasmussen [2008]. MCMC stands out with the highest posterior fidelity at prohibitive cost, and has been considered intractable for large data until recently. SVGP is later developed as the dominant scalable approach to GP inference with non-Gaussian likelihoods, which combines the inducing-variable framework of sparse GPs [Quiñonero-Candela and Rasmussen, 2005, Titsias, 2009] with stochastic variational inference [Hensman et al., 2013]. For classification, Hensman et al. [2015] used mini-batches to optimize variational lower bound and the inducing points, approximating the posterior over latent function through a tractable Gaussian variational approximation. More generally, variational inference based on the reverse KL objective can underestimate posterior variance [Blei et al., 2017, Yao et al., 2018]. This loss of posterior uncertainty is especially important in safety-critical classification problems, where miscalibrated probabilities and overconfidence under distribution shift can have direct downstream consequences [Guo et al., 2017, Ovadia et al., 2019]. We use SVGP classification [Hensman et al., 2015] as our primary scalable GP baseline.

Neural Networks. For neural network baselines, we use Compared to multiple architecture we used three architectures that are commonly used in histopathology classification. We use Residual MLP as head with foundation models like Phikon [Filiot et al., 2023] which help learn frozen level embeddings on $\sim 40M$ of images. We also compare against U-Net [Ronneberger et al., 2015], a standard convolutional network for medical imaging. Additionally, ViTs [Dosovitskiy et al., 2021] are proven to show competitive results on image related task when paired with large scale pretraining.

3 Method

3.1 Low-rank Factorization: Accelerated RPCholesky

The accelerated RPCholesky factors the dense $N \times N$ kernel matrix into low rank factor $\mathbf{L} \in \mathbb{R}^{N \times k}$ such that $\mathbf{K} \approx \mathbf{L}\mathbf{L}^T$. Using the AbstractPSDMatrix class defined in Epperly et al. [2025], the low-rank factor is computed directly from the training data, while the actual $N \times N$ kernel matrix is never materialized. The input is an $N \times d$ matrix and the output is a $N \times k$ matrix $\mathbf{L}$ with low rank k .

3.2 Posterior Sampling: Hamiltonian Monte Carlo (HMC) with non-centered parameterization (NCP)

As mentioned above, the posterior of GPC $p(\mathbf{f}|\mathbf{y}, \mathbf{X}, \theta)$ is not tractable. SVGP attempts to solve it by assuming the posterior is Gaussian and perform stochastic variational inference. An alternative for sampling this intractable posterior would be MCMC samplers. The paper Betancourt and Girolami [2013] inspired our design.

GPC has a difficult posterior geometry with $\mathbf{f}$ inherits strong correlations from the kernel matrix $\mathbf{K}_\theta$, and naive HMC on geometries with difficult curvatures is inefficient. We can use non-centered parameterization to solve this problem as follows: Set $\boldsymbol{\nu} \sim \mathcal{N}(0, I)$ and

$$\mathbf{f} = \mathbf{L}_\theta \boldsymbol{\nu}, \quad (1)$$

where $\boldsymbol{\nu} \in \mathbb{R}^k$ is the HMC-sampled latent function after non-centered parameterization, and $\mathbf{f} = \mathbf{L}_\theta \boldsymbol{\nu}$ is the actual training latent function. We now have an isotropic prior geometry, no longer sampling

from $\mathbf{f} \sim \mathcal{N}(0, \mathbf{K}_\theta)$ which has possibly extreme eigenvalue spread. After NCP, we use Euclidean HMC which the paper suggests. This means that

$$\rho \sim \mathcal{N}(0, \mathbf{I}), K(\rho) = \frac{1}{2} \rho^T \rho.$$

This paper gives us the theoretical foundation for using HMC as the MCMC sampler for GPC, but we plan to conduct experiments comparing the performance of different MCMC samplers on GPC.

3.3 Pivots as Inducing Points

Computing the predictive distribution is also challenging at large scale. For the joint Gaussian prior over training and test latent values, $\mathbf{f}$ and $\mathbf{f}_*$, exact GP prediction requires solving linear systems involving the $N \times N$ training kernel matrix. This costs $O(N^3)$ time for a dense factorization and $O(N^2)$ memory, which is infeasible for large N . We therefore use the RPCholesky pivots as inducing points, adapting ideas from sparse inducing-variable approximations.

We adapt the inducing-variable method, specifically the deterministic training conditional (DTC) approximation [Quiñero-Candela and Rasmussen, 2005]. Let $\mathcal{P} \subset \{1, \dots, N\}$ denote the set of k RPCholesky pivot indices, and let

$$\mathbf{u} = \mathbf{f}_{\mathcal{P}} \in \mathbb{R}^k$$

be the latent function values at these pivot points. Under the DTC conditional independence assumption

$$\mathbf{f}_* \perp \mathbf{f} \mid \mathbf{u},$$

prediction depends on the training data only through the k -dimensional inducing variable $\mathbf{u}$, reducing the dominant predictive linear algebra from N -dimensional to k -dimensional computations.

Unlike standard inducing-variable methods that optimize inducing locations during training [Snelson and Ghahramani, 2005, Titsias, 2009], we fix the inducing points to be the RPCholesky pivots $\mathcal{P}$. This choice is natural in our setting because the pivots are already selected by the low-rank kernel approximation procedure and correspond to informative columns of the kernel matrix [Chen et al., 2025]. The corresponding inducing variables satisfy

$$\mathbf{u} = \mathbf{f}_{\mathcal{P}} \sim \mathcal{N}(0, \mathbf{K}_{\mathcal{P}, \mathcal{P}}).$$

Moreover, the RPCholesky factor $\mathbf{L} \in \mathbb{R}^{N \times k}$ satisfies

$$\mathbf{L}[\mathcal{P}, :] \mathbf{L}[\mathcal{P}, :]^T = \mathbf{K}_{\mathcal{P}, \mathcal{P}}, \quad (2)$$

so $\mathbf{K}_{\mathcal{P}, \mathcal{P}}$ is available directly from the precomputed low-rank factor. Therefore, for an HMC sample $\boldsymbol{\nu}$ with $\mathbf{f} = \mathbf{L}\boldsymbol{\nu}$, the corresponding inducing variable is

$$\mathbf{u} = \mathbf{f}_{\mathcal{P}} = \mathbf{L}[\mathcal{P}, :] \boldsymbol{\nu}.$$

For a test set $\mathbf{X}_*$, the DTC predictive conditional is

$$p(\mathbf{f}_* \mid \mathbf{f}_{\mathcal{P}}, \mathbf{X}_*) = \mathcal{N}\left(\mathbf{K}_{*, \mathcal{P}} \mathbf{K}_{\mathcal{P}, \mathcal{P}}^{-1} \mathbf{f}_{\mathcal{P}}, \mathbf{K}_{*,*} - \mathbf{K}_{*, \mathcal{P}} \mathbf{K}_{\mathcal{P}, \mathcal{P}}^{-1} \mathbf{K}_{\mathcal{P},*}\right). \quad (3)$$

Here, $\mathbf{K}_{*, \mathcal{P}} = k(\mathbf{X}_*, \mathbf{X}_{\mathcal{P}})$ is the test–pivot cross-kernel matrix, $\mathbf{K}_{\mathcal{P},*} = \mathbf{K}_{*, \mathcal{P}}^T$, and $\mathbf{K}_{*,*} = k(\mathbf{X}_*, \mathbf{X}_*)$ is the test–test kernel matrix. In practice, for marginal class probabilities we only need the diagonal of the conditional covariance, so $\mathbf{K}_{*,*}$ can be replaced by its diagonal.

The resulting predictive cost is

$$O(k^3 + n_{\text{test}} k^2 + n_{\text{test}} k d),$$

where the $O(k^3)$ term comes from factorizing $\mathbf{K}_{\mathcal{P}, \mathcal{P}}$, the $O(n_{\text{test}} k^2)$ term comes from applying $\mathbf{K}_{\mathcal{P}, \mathcal{P}}^{-1}$ to the test–pivot kernel matrix, and the $O(n_{\text{test}} k d)$ term comes from forming $\mathbf{K}_{*, \mathcal{P}}$. Since in our experiments $k = O(10^2)$ and $n_{\text{test}} = O(10^4)$, the $n_{\text{test}} k^2$ term dominates.

3.4 Computing the predictive

The class probability for a test point is

$$p(\mathbf{y}^* | \mathbf{y}, \mathbf{X}^*) = \mathbb{E}_{p(\mathbf{f}^* | \mathbf{y}, \mathbf{X}^*)} [\sigma(\mathbf{f}^*)] = \int \sigma(\mathbf{f}^*) p(\mathbf{f}^* | \mathbf{y}, \mathbf{X}^*) d\mathbf{f}^* \quad (4)$$

for $\mathbf{y}$ the training label, $\mathbf{X}^*$ the testing data, σ the sigmoid function. The integral is approximated using S HMC samples and R conditional draws per sample:

$$\hat{p}(y^* = 1 | \mathbf{y}, \mathbf{x}^*) = \frac{1}{SR} \sum_{s=1}^S \sum_{r=1}^R \sigma(f_*^{(s,r)}), \quad (5)$$

where $f_*^{(s,r)} \sim p(f^* | \mathbf{u}^{(s)}, \mathbf{x}^*)$ and $\mathbf{u}^{(s)} = \mathbf{L}[\mathcal{P}, :] \boldsymbol{\nu}^{(s)}$ is constructed from the s -th HMC sample $\boldsymbol{\nu}^{(s)}$. This yields a probability instead of a binary label, but it does give us various important metrics.

4 Experiments

4.1 Datasets and Feature Extraction

We evaluate on two histopathology benchmarks for identifying breast cancer metastasis. **PCam** (Veeling et al. [2018]) contains 327,680 colored image patches (96×96) divided into 262,144 training, 32,768 validation, and 32,768 test images, with binary labels indicating the presence of tumor tissue.

We also use **CAMELYON17-WILDS** (Bandi et al. [2019]), which is the **CAMELYON17-WILDS** dataset accessed via the WILDS benchmark (Koh et al. [2021]). It contains 455,954 96×96 patches divided into 302,000 training images from 3 hospitals, 34,000 valid images from the fourth hospital, and 85,000 testing images from the fifth hospital. Both are standard benchmarks for cancer diagnosis where calibrated uncertainty is relevant. In addition, **CAMELYON17-WILDS** tests the method’s calibration under distribution shift.

Embedding as features The training datasets both have images of size 96×96 pixels. Although ideally we can run GPC over raw pixels, the kernel matrices are not informative in such high dimensions. Thus, we extract 512 features using *Phikon*, a ViT-Base transformer pretrained on $\sim 40\text{M}$ histopathology data.

4.2 Experiment 1: Initial Explorations

Experiment 1 justifies the methodological choices we use in Experiment 2. There are three sub-experiments, all using small scale data ($N = 2000$ training points). Experiments 1a and 1b use synthetic data: two Gaussian blobs centered on either side of the y -axis, labeled 0 and 1 respectively. Experiment 1c uses a random subsample from **PCam**.

Experiment 1a: Speedup from accelerated RPCholesky in GPC First, we verify that accelerated RPCholesky not only reduces computational cost for matrix factorization, but actual GPC as well. For the kernel matrix, we compute and compare HMC sampling results based on equation (??) with $\mathbf{L}$ the dense $N \times N$ matrix directly, or its low rank approximation with rank $k = \{20, 50, 100\}$. Using Random-Walk Metropolis (RWM) as a baseline sampler, we measure factor computation time and total time. We repeat the run for 10 times and calculate the average and mean. Both the factor time and total time increase as k increases, but not at the theoretical $O(k^2 N)$ scale due to noise. With small std, the factor time has a 6x improvement, and the total time has a 15x improvement.

Experiment 1b: Comparison of MCMC samplers on synthetic data After establishing the superiority of low-rank methods, we use the same synthetic data and compare three MCMC samplers on the low-rank GPC posterior: RWM, the Metropolis-adjusted Langevin algorithm (MALA), and HMC. We use accelerated RPCholesky for kernel matrix factorization with fixed rank $k = 200$. The metrics we evaluate are: autocorrelation time τ , effective sample size (ESS), and ESS per second. Higher ESS per second means that we are taking more independent MCMC samples per second. MALA and HMC clearly outperforms RWM in terms of smaller τ , higher ESS, and higher ESS/sec. Unfortunately, MALA and HMC don’t exhibit big difference in ESS/sec.

Experiment 1c: Sampler comparison on PCam While theoretically HMC has better performance than MALA, the results in Experiment 1b don’t fully support that. We suspect that the synthetic data’s posterior is not complicated enough to show true difference between MCMC samplers. Thus, we repeat the sampler comparison on a random 2000-point subsample of **PCam**, with the same RPCholesky rank ($k = 200$) and metrics. In actual data, HMC establishes dominating advantage over the other two MCMC samplers.

4.3 Experiment 2: Full-scale HMC GPC

We scale method to the full training set of **PCam** and **CAMELYON17-WILDS**, and compare against Stochastic Variational Gaussian Process (SVGP) classification (Hensman et al. [2015]), the standard GP baseline.

SVGP baseline. We use stochastic variational Gaussian process classification (SVGP) as the main scalable GP baseline. We implement the baseline in GPyTorch. The SVGP model is trained using stochastic gradient optimization, and we report the best full-scale non-standardized result for each dataset.

Experiment setup For our method, we factor the $N \times N$ training kernel matrix using accelerated RPCholesky, where N is the number of training examples in each dataset. We then sample the low-rank GP posterior using the non-centered parameterization in eq. (1). For the HMC sampler, we use leapfrog integration with step-size adaptation via dual averaging [Hoffman and Gelman, 2014], following standard HMC practice [Betancourt, 2017]. We set the number of warmup steps to $n_{\text{warmup}} = 1000$ and retain $n_{\text{sample}} = 1000$ post-warmup samples. We perform a sequential parameter sweep on optimizing the leap frog steps n_{leapfrog} , rank k , and bandwidth l . Then we run the experiment on the full training data with parameters $n_{\text{leapfrog}} = 25$, $k = 200$, and $l = 49$.

For the SVGP baseline, we use bandwidth $l = 20$ for initialization. The number of inducing points is fixed to $M = 200$ for both **PCam** and **CAMELYON17-WILDS**, while the inducing locations are initialized from the training embeddings and optimized jointly with the variational parameters during training.

Metrics We report metrics covering classification performance, calibration, probabilistic predictive quality, and computational cost: accuracy, negative log-likelihood (NLL), expected calibration error (ECE), area under the receiver operating characteristic curve (AUROC), Brier score.

4.4 Experiment 3: Neural Network Methods

Identifying the best NN-based baseline. We evaluate different NN-based methods on metrics mentioned in Experiment 2 and identify best neural network model to compare with our proposed method. We considered 3 different methods and tested them with both **PCam** and **CAMELYON17-WILDS**. The three models are (1) **Phikon with a standard prediction head**: we use 512-length Phikon embeddings of the images to train a residual MLP; (2) **U-Net classifier**: we train a U-Net classifier directly on the raw pixel values; (3) **ViT-S trained from scratch**: we train a ViT-Small model directly on the raw pixel data.

Based on our experiments, U-Net achieves the strongest performance among the three neural baselines. We hypothesize that ViT-Small underperforms due to its high data requirements when training from scratch, while the Phikon-based method might underperform due to our dataset being out-of-distribution for Phikon’s training set resulting in suboptimal representation.

4.5 Results

Computational Cost All of the experiments are run on High-Performance Computing (HPC) cluster and on a single GPU. Experiment 2 runs in approximately 22 minutes for **PCam**: Only 16 seconds for accelerated RPCholesky kernel factorization, and ~ 20 minutes for HMC sampling and ~ 2 seconds for predictive inference. By comparison, a dense Cholesky factorization for $N \approx 300,000$ requires 550 GB storage and can’t be computed unless using special infrastructure. The training time, not including the embedding generation time, is ~ 6 minutes for SVGP, and U-net is around an hour, which means that HMC GPC isn’t the best in terms of time, but comparable.

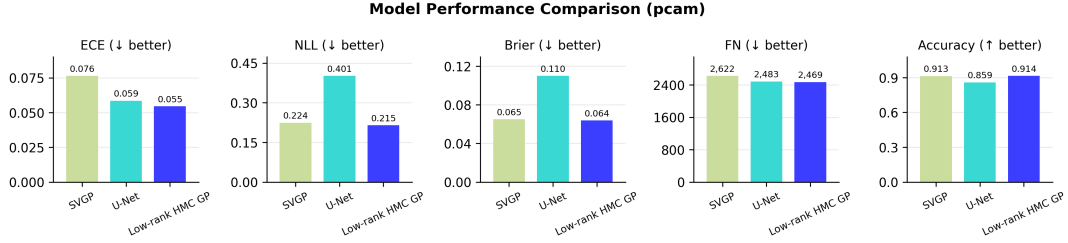


Figure 1: Final PCam comparison across SVGP, U-Net, and our method (Low-rank HMC GP).

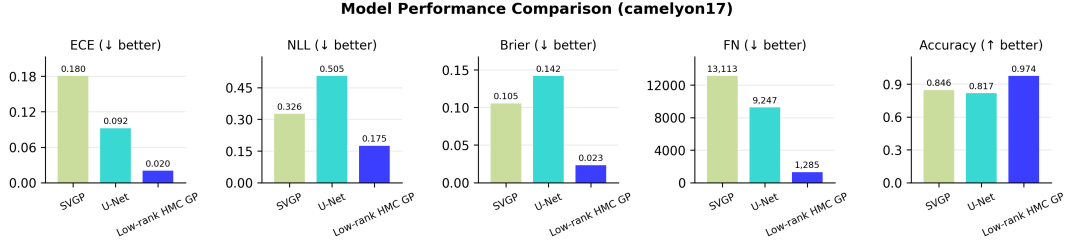


Figure 2: Final CAMELYON17 comparison across SVGP, U-Net, and our method (Low-rank HMC GP).

Data Analysis Figures 1 and 2 compare model performance on **PCam** and **CAMELYON17-WILDS**. Our low-rank HMC GP improves most of the metrics against the SVGP and NN-based baselines across both datasets. We are seeing bigger performance gain over **CAMELYON17-WILDS**, where the test set is generated from data at a held-out hospital. For all probabilistic metrics our proposed method is exhibiting impressive gain; Based on results mentioned in Table 1 reports the PCam results, while Table 2 reports the CAMELYON17 results. over SVGP baseline, our method reduces ECE by 87.3%, NLL by 44.5%, Brier score by 76.8%, and false negatives by 89.4%; over NN baseline, our method reduces ECE by 78.2%, NLL by 65.4%, Brier score by 83.6%, and false negatives by 86.1%. This suggests that our method is less overconfident and less likely to be confidently wrong, and most importantly less likely to misdiagnose cancer-positive samples.

Calibration under distribution shift In **PCam**, the training, validation, and test images are drawn from the same hospital, so there is no data distribution shift in **PCam**. In contrast, **CAMELYON17-WILDS** consists of training set images from three hospitals, validated on a fourth hospital, and tested on a fifth. It shows that distribution shift has a clear effect on model calibration and predictive quality.

We notice that moving from **PCam** to **CAMELYON17-WILDS**, the SVGP and U-Net baselines degrade under data distribution shift. This suggests that HMC-based posterior uncertainty provides stronger calibration under data distribution differs from the training distribution.

5 Discussion

From the experiments, we saw that our proposed methods improves probabilistic metrics including ECE, NLL, Brier, and false negative over both SVGP and NN baselines. These improvements are the most apparent on the **CAMELYON17-WILDS** dataset. It is worth noting that the **CAMELYON17-WILDS** dataset was specifically created with train and test sets coming from different hospitals. This suggests that posterior sampling deployed in our method is particularly effective under distribution shift.

While our method still takes longer to train compared to the SVGP and NN baselines, we confirm that applying Accelerated RPCholesky make MCMC-based GPC feasible for modern, large dataset rather than purely theoretical. The primary bottleneck has now shifted to HMC warmup and sampling, which takes several minutes. Future work could explore optimization in the sampling stage such as more efficient HMC tuning, parallel chains, or lower-variance posterior approximations with the goal

Table 1: PCam predictive metrics. Improvements are relative changes of Low-rank HMC GP against each baseline; lower is better for ECE, NLL, Brier, FNR, FPR, FN, and FP.

Metric	SVGP	U-Net	Low-rank HMC GP	Gain vs SVGP	Gain vs U-Net
ECE	0.101	0.059	0.055	+46.2%	+6.9%
NLL	0.287	0.401	0.215	+25.2%	+46.5%
Brier	0.087	0.110	0.064	+26.3%	+41.9%
Accuracy	0.882	0.859	0.914	+3.7%	+6.4%
AUROC	0.971	0.925	0.978	+0.7%	+5.7%

Table 2: CAMELYON17 predictive metrics. Improvements are relative changes of Low-rank HMC GP against each baseline; lower is better for ECE, NLL, Brier, FNR, FPR, FN, and FP.

Metric	SVGP	U-Net	Low-rank HMC GP	Gain vs SVGP	Gain vs U-Net
ECE	0.158	0.092	0.020	+87.3%	+78.2%
NLL	0.315	0.505	0.175	+44.5%	+65.4%
Brier	0.100	0.142	0.023	+76.8%	+83.6%
Accuracy	0.858	0.817	0.974	+13.5%	+19.2%
AUROC	0.990	0.893	0.992	+0.2%	+11.1%

of reducing runtime further. Further work on finding a strong NN-based baseline has been planned, and we are also interested in testing the proposed method on other tasks in histopathology such as cancer sub-typing.

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A Technical References

A.1 Gaussian Process Classification

Gaussian Process Classification (GPC) is a hierarchical Bayesian model with three levels: A kernel hyperparameter prior, a latent function with a GP prior, and the Bernoulli mapping that maps that latent function to binary labels.

1. Hyperparameters: θ : The hyperparameter $\theta = (\sigma_f, l)$ for the kernel matrix with prior $p(\theta)$. The matrix entries are calculated using the kernel function:

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2l^2}\right). \quad (6)$$

The kernel matrix is semi-positive definite. In our case, we fix θ while sampling $\mathbf{f}$, and do parameter sweeps later to find a good θ .

2. Latent function $\mathbf{f}|\theta, \mathbf{X} \sim \mathcal{N}(0, \mathbf{K}_\theta(\mathbf{X}, \mathbf{X}))$;
3. $y_i|f_i \sim \text{Bernoulli}(\sigma(f_i))$: This is independent from θ . We choose to use sigmoid as the link function.

A.2 Inference Problem

When $\mathbf{f}, \theta$ are jointly sampled, the posterior is

$$p(\mathbf{f}, \theta|\mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y}|\mathbf{f}) \cdot p(\mathbf{f}|\theta, \mathbf{X}) \cdot p(\theta)}{p(\mathbf{y}|\mathbf{X})} \quad (7)$$

In our work, we sample $\mathbf{f}$ while fixing $\theta = (1, l)$ for some fixed l , so the posterior we are sampling is

$$p(\mathbf{f}|\mathbf{y}, \mathbf{X}, \theta) = \frac{p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\theta, \mathbf{X})}{p(\mathbf{y}|\theta, \mathbf{X})} \quad (8)$$

We fix the variance at $\sigma_f^2 = 1$ which is consistent to the accelerated RPCholesky paper. The posterior of the latent function $p(\mathbf{f}|\mathbf{y}, \mathbf{X}, \theta)$ is generally not tractable since it is proportional to a Bernoulli likelihood times a Gaussian prior. In addition, The kernel matrix is an $N \times N$ matrix, which costs $O(N^2)$ memory to store, and $O(N^3)$ time to materialize. For even the smallest datasets among the three that we choose, **PCam**, $N = 262, 144$, the matrix computation would be intractable. However, we will show how to make fully Bayesian GPC tractable.